

FRANCESCO BARILE

CONTACT INFORMATION

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ACADEMIC POSITIONS

University of Milano-Bicocca, Milan, Italy

• **Postdoctoral Research Fellow**

Mar 2025 - present

(*Assegno di Ricerca* pursuant to Article 22 of Italian Law No. 240/2010)

Department of Economics, Management and Statistics

Project: Joint Bayesian Learning of Related Clinical Studies [\[link\]](#)

Project PI: Prof. Bernardo Nipoti, University of Milano-Bicocca

Partner unit coordinator: Prof. Francesco C. Stingo, University of Florence

Funding: Fondazione Cariplo

EDUCATION

University of Milano-Bicocca, Milan, Italy

• **Ph.D. in Statistics**

2020 - 2024

Department of Economics, Management and Statistics

Defense date: Feb 14th, 2025

Thesis title: “Advances in Bayesian Methods:

Final evaluation: Excellent

Nonparametric Network Modeling and Scalable Computations”

Supervisor: Prof. Bernardo Nipoti

University of Calabria, Rende, Italy

• **M.Sc. in Finance and Insurance** (English)

2016 - 2019

Department of Economics, Statistics and Finance

Thesis title: “Estimating Stochastic Models for commodity futures”

110/110 cum laude

Supervisor: Prof. Arturo Leccadito

• **B.Sc. in Statistics for Business and Insurance**

2012 - 2015

Department of Economics, Statistics and Finance

Thesis title: “Test di Wilcoxon: applicazioni in ambito medico-farmaceutico”

110/110 cum laude

Supervisor: Prof. Agostino Tarsitano

RESEARCH VISITS

• **Visiting Researcher**

Nov 2023 - Aug 2024

Lancaster University, Lancaster, United Kingdom

School of Mathematical Sciences

Supervisor: Prof. Christopher Nemeth

• **Visiting Graduate Student (Research Only)**

Aug - Dec 2018

University of Western Ontario, London, Canada

Department of Statistical and Actuarial Sciences

Supervisor: Prof. Rogemar Mamon

Research topic: Statistical modeling of lumber price dynamics

RESEARCH INTERESTS

Bayesian nonparametrics, Network modeling, Computational statistics, Bayesian statistics

SCHOLARSHIPS AND AWARDS

● Junior Travel Support Award ISBA 2026, Nagoya, Japan	2026
● Junior Travel Support Award BayesComp 2025, Singapore	2025
● Exchange Extra-UE Funded Mobility Programme Award (10 months) University of Milano-Bicocca	2023
● Ph.D. Scholarship Award (4 years) University of Milano-Bicocca	2020
● Overseas Research Scholarship Award (4 months) University of Calabria	2018

PUBLICATIONS

REFERRED JOURNAL ARTICLES

- [1] **Barile, F.**, Lunagómez, S., & Nipoti, B. (2025). Bayesian nonparametric modeling of heterogeneous populations of networks. *Bayesian Analysis*. Advance Publication. doi: [10.1214/26-BA1588](https://doi.org/10.1214/26-BA1588).

REFERRED CONFERENCE PROCEEDINGS

- [1] **Barile, F.**, Lunagómez, S., & Nipoti, B. (2023) Flexible modelling of heterogeneous populations of networks: a Bayesian nonparametric approach. In C. Rampichini, M. La Rocca, P. Coretto, G. Giordano, M.L. Parrella (Eds.). *Book of Abstracts and Short Papers CLADAG 2023* (p. 69). Pearson. ISBN: 978-88-9193-563-2. [\[link\]](#)
- [2] **Barile, F.**, Lunagómez, S., & Nipoti, B. (2025) Clustering multiple network data with an application to the EU food trade market. In: A. Pollice, P. Mariani (Eds.). *Methodological and Applied Statistics and Demography II: SIS 2024, Short Papers, Solicited Sessions* (pp. 68–72). Springer. ISBN: 978-3-031-64350-7. doi: [10.1007/978-3-031-64350-7](https://doi.org/10.1007/978-3-031-64350-7).
- [3] **Barile, F.**, Dallari, S., Pacifici, C., & Grün, B. (2025). Assessing neuron response to external stimuli with a data-driven procedure for spike train extraction and GAMLSS regressions. In: A. Canale, A. Luati, S. Mazzuco, R. Piccarreta, N. Sartori, P. Secchi (Eds.). *Advances in Neural Data Science* (pp. 35–55). Springer. ISBN: 978-3-031-70638-7. doi: [10.1007/978-3-031-70638-7_3](https://doi.org/10.1007/978-3-031-70638-7_3).

MANUSCRIPT UNDER REVIEW

- [1] **Barile, F.** & Nemeth, C. (2024). Control variate-based stochastic sampling from the probability simplex. *Under review (2026)*. *ArXiv*: [\[link\]](#). *Code*: [\[link\]](#)
- [2] **Barile, F.**, Capozio, S. & Nipoti, B. (2026). Efficient Bayesian inference for multiple network data. *Under review*. *Code*: [\[link\]](#)

MANUSCRIPT IN PREPARATION

- [1] **Barile, F.**, Borrotti, M., Nipoti, B., Poli, G. & Stingo, F. C. (2026+). Data Integration and Missing Imputation for Joint Bayesian Learning under Multiple Nonignorable Missing Mechanism. *In preparation*.
- [2] **Barile, F.**, Nieto-Barajas, L.E., & Nipoti, B. (2026+). A Network of Dependent Dirichlet Processes. *In preparation*.

- [3] **Barile, F.**, Lunagómez, S., & Nemeth, C. (2026+). Covariate-dependent modeling of hypergraph data. *In preparation*.

PH.D. THESIS

- [1] **Barile, F.** (2025). Advances in Bayesian Methods: Nonparametric Network Modeling and Scalable Computations. Ph.D. thesis, XXXVI cycle [\[link\]](#)

PRESENTATIONS AND SEMINARS

INVITED PRESENTATIONS

- CFE-CMStatistics 2026, Dec 12-14, 2026
HTW Berlin, University of Applied Sciences, Berlin, Germany
- StaTalk 2025, University of Milano-Bicocca, Milan, Italy Jun 13-14, 2025
- ECOSTA 2023, Waseda University, Tokyo, Japan Aug 1-3, 2023

CONTRIBUTED PRESENTATIONS

- BAYSM 2023, Online mode Nov 13-17, 2023

POSTERS

- ISBA 2026, Nagoya, Japan Jun 28 - Jul 3, 2026
- Bayes Comp 2025, Singapore Jun 16-20, 2025
- SISBayes, Padova, Italy Sep 4-5, 2025
- Complex Graphical Models for Biological Network Science, Padova, Italy Sep 4, 2024
- ISBA 2024, Venice, Italy Jul 1-7, 2024
- IMS ICSDS 2022, Florence, Italy Dec 13-16, 2022

SEMINARS

- Seminar in the Computational Statistics and Machine Learning series Feb 2024
School of Mathematical Sciences, Lancaster University, Lancaster, UK

WORKSHOPS

- Data Research Camp 2022, VIU, San Servolo Island, Venice, Italy Jul 12-15, 2022

TEACHING

Lecturer

- Inferential Statistics [ENG] 2026/2027
(Master's degree in International Economics and Master's degree in Global Management)
Department of Economics, Management and Statistics, University of Milano-Bicocca

Adjunct Lecturer

- Industrial Economics and Dynamics [ITA] 2021/2022
(Master's degree in Statistical and Economic Sciences)
Department of Economics, Management and Statistics, University of Milano-Bicocca

Teaching Assistant

- Inferential Statistics [ENG] (Master's degree in International Economics) 2025/2026
Department of Economics, Management and Statistics, University of Milano-Bicocca
- Inferential Statistics [ENG] (Master's degree in International Economics) 2024/2025
Department of Economics, Management and Statistics, University of Milano-Bicocca
- Advanced Statistics [ENG] (Master's degree in Marketing and Global Markets) 2024/2025
Department of Economics, Management and Statistics, University of Milano-Bicocca
- Inferential Statistics [ENG] (Master's degree in International Economics) 2023/2024
Department of Economics, Management and Statistics, University of Milano-Bicocca
- Advanced Statistics [ENG] (Master's degree in Marketing and Global Markets) 2023/2024
Department of Economics, Management and Statistics, University of Milano-Bicocca
- Inferential Statistics [ENG] (Master's degree in International Economics) 2022/2023
Department of Economics, Management and Statistics, University of Milano-Bicocca

SERVICE TO PROFESSION

ORGANIZATION OF SCIENTIFIC EVENTS

Organizer and chair of the Invited Session “Recent advances in Bayesian variable selection and shrinkage” of CFE-CMStatistics 2026 [[link](#)]

REFeree SERVICE

Computational Statistics and Data Analysis; Communications in Statistics - Simulation and Computation; IEEE Journal of Biomedical and Health Informatics.

POSITIONS IN ACADEMIC SOCIETIES

Member of ISBA, j-ISBA Section, BNP-ISBA Section

Member of the International Society for Bayesian Analysis (ISBA), Junior ISBA Section, Bayesian non-parametric ISBA Section

Member of SIS, y-SIS

Member of the Italian Statistical Society (Società Italiana di Statistica, SIS), Young SIS

THESIS SUPERVISION

Co-supervision of a Master's thesis, Master's Degree in Statistical and Economic Sciences, Department of Economics, Management and Statistics, University of Milano-Bicocca

COMPUTER SKILLS

Programming languages R, C++, Python (basic), SQL, Java, L^AT_EX
Software SAS, Stata
RStudio:
• C++ based packages creation;
• parallel programming;
• create dynamic reports and reproducible research using L^AT_EX;
• project version control via GitHub.

LANGUAGES

Italian Mother tongue
English Fluent
French Basic

OTHER EXPERIENCES

• Junior Data Scientist CRIF S.p.A.	Apr 2019 - Sep 2020
• Degree Program Tutor Department of Economics, Statistics and Finance, University of Calabria	Feb - Jul 2018
• Degree Program Tutor Department of Economics, Statistics and Finance, University of Calabria	Apr - Oct 2017

The statements contained in this curriculum vitae are made pursuant to Articles 46 and 47 of Presidential Decree No. 445/2000.

Milan, September 23, 2026